

The role of tongue root anteriority in the Dolgan vowel system

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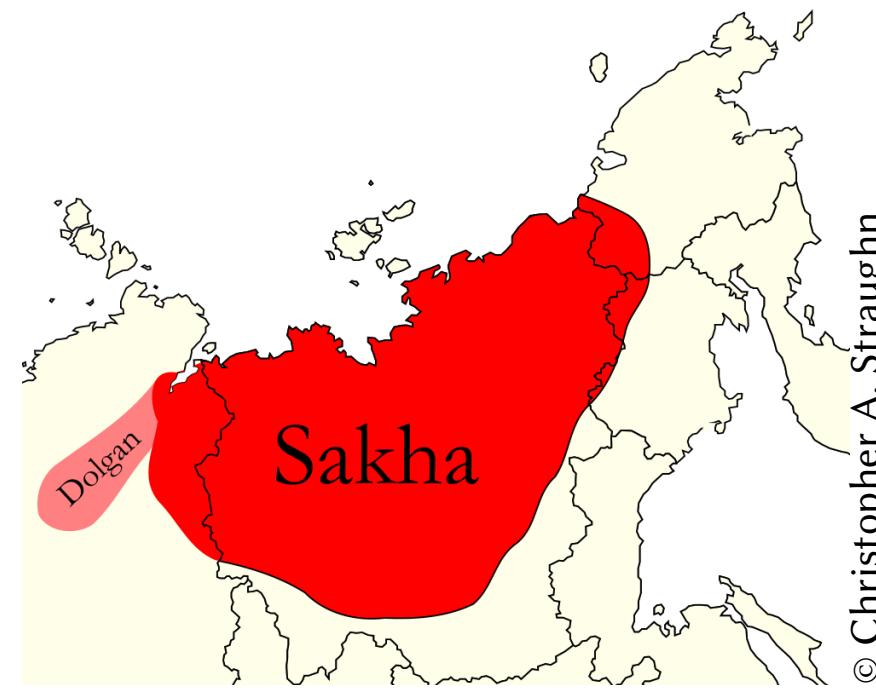
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Dolgan

- Lena Turkic language, spoken on Taymyr peninsula (Arctic Siberia)
- Endangered, ~5k speakers, loss of intergenerational transmission



Phonological evidence: alternation of /G/

/G/: surfaces as **velar** or **uvular** depending on preceding vowel

Kyrgyz (&c.)

	front	back
+high	[ini-ye] ‘brother-DAT’	[a:ɾu-βa] ‘bee-DAT’
-high	[ene-ye] ‘mother-DAT’	[ɑɾa-βa] ‘saw-DAT’

- **uvular G** / V_[+back] → **velar G** / V_[-back] →
- ↳ V backness associated with TR backness
- coproduced: TB backness & TR backness (Washington, 2016, 2019)

Sakha

+high	[iti:-gɛ] ‘heat-DAT’	[ɑɾu:-ga] ‘century-DAT’
-high	[y]ɛ-βɛ] ‘century-DAT’	[taba-βa] ‘reindeer-DAT’

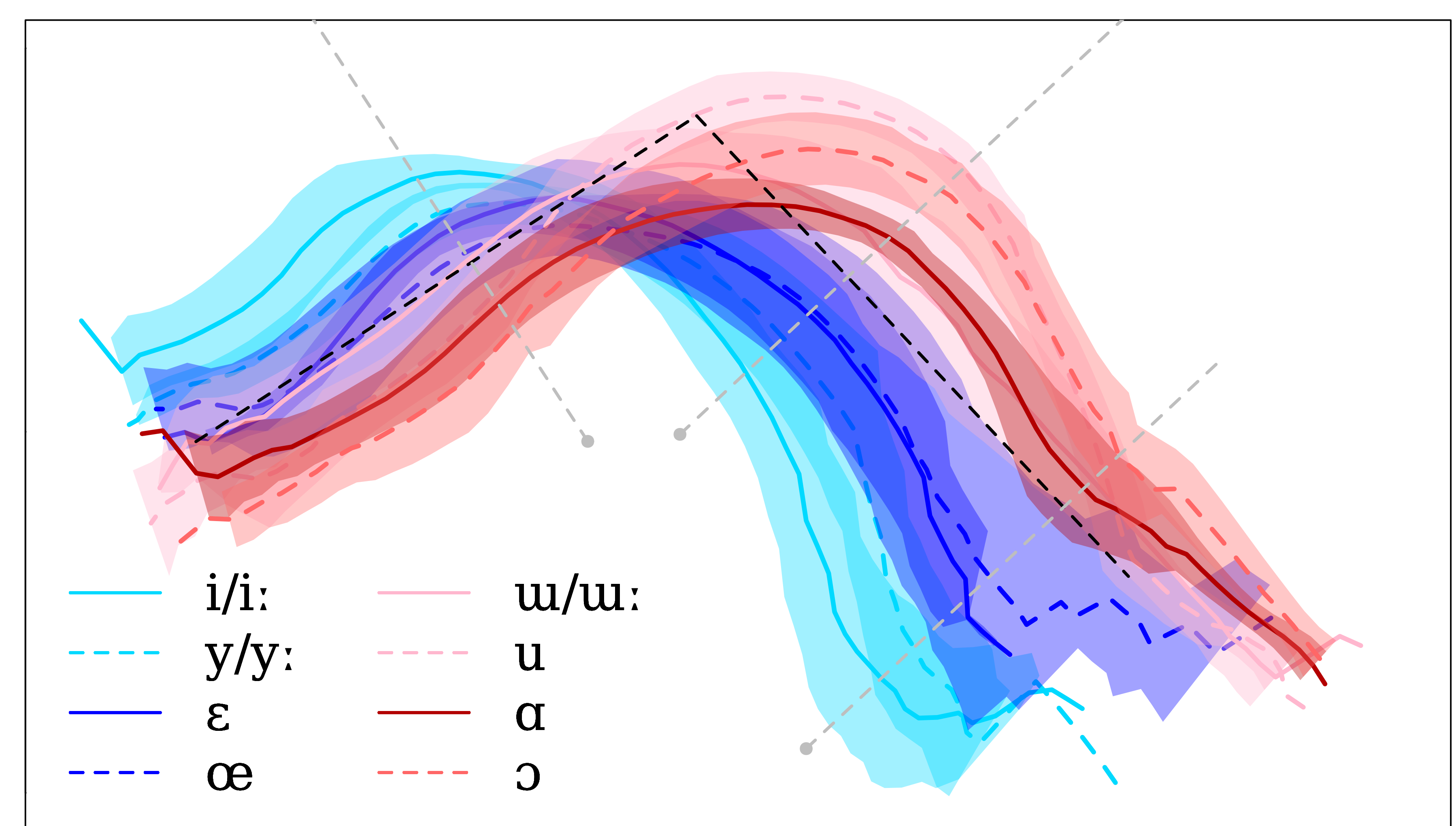
- **uvular G** / V_[-high] → **velar G** / V_[+high] →
- ↳ V height associated with TR backness
- coproduced: TB height & TR backness (Vaux, 2009)

Dolgan

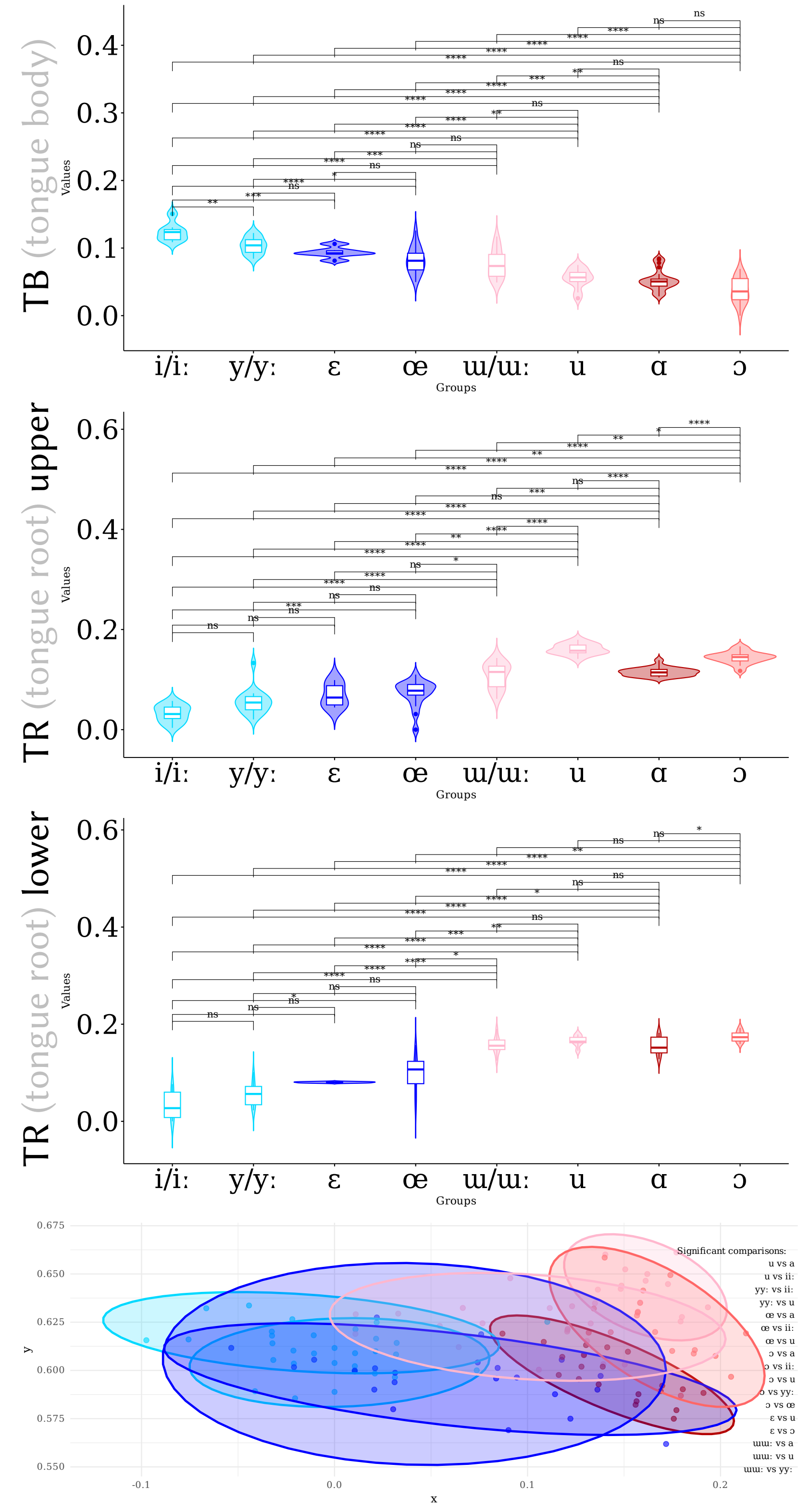
+high	[iti:-gɛ] ‘heat-DAT’	[ɑɾu:-ga] ‘century-DAT’
-high	[yjɛ-gɛ] ‘time-DAT’	[taba-βa] ‘reindeer-DAT’

- **uvular G** / V_[-hi,+bk] → **velar G** / elsewhere (Убрятова, 1985; Däbritz, 2022)
- ↳ V height & backness associated w. TR backness
- coproduced: TB height & TB & TR backness!

Results



Significance (paired t-tests)



		TB		TR		
		pair	middle	PMC	upper	lower
height	i-ε	0.00052	1.00000	0.28013	0.09608	
	y-œ	0.02035	1.00000	1.00000	0.19771	
	u-ɑ	0.00555	0.00049	1.00000	1.00000	
	u-ɔ	0.16815	0.01150	0.02140	1.00000	
rounding	i-u	1.6e-07	0.20473	3.5e-08	2.4e-05	
	y-u	1.3e-07	4.3e-13	3.8e-13	1.2e-07	
	ε-ɑ	5.0e-06	1.00000	0.11347	1.8e-09	
	œ-ɔ	2.5e-06	4.2e-10	2.7e-08	0.00252	
backness	i-y	0.00732	0.00584	0.12860	1.00000	
	ε-œ	1.00000	1.00000	1.00000	1.00000	
	ɔ-ɑ	0.27664	1.3e-06	4.3e-07	0.03711	
	u-u	0.07095	1.4e-06	1.4e-05	1.00000	
p values: ns ≥ 0.05 > * ≥ 0.01 > ** ≥ 0.001 > *** ≥ 0.0001 ≥ ****						

Methodology

- Recorded** L1 speaker of Lower Xatanga Dolgan (author 4)
- Wordlist task, carrier phrase Зо́я __ ди́ир ‘Zoya says __’ (in custom orthography)
- Articulate Instruments’ Micro ultrasound system, AAA software (Wrench, 2017)
- Annotated** midsagittal tongue surface using UltraTrace (Murphy et al., 2020)
- frame closest to midpoint of V in 1st syllable of each word
- Plotted** and **analysed** using ultrapolaRplot (Outkin & Washington, 2024)
- Perpendicular rays intersect various areas of the tongue (mid TB, upper TR, etc.)
- Pairwise t-test at intersections for significance (adjusted for Type I error)
- Pairwise t-test compares maximum curvature (PMC) groups by calculating differences in distance to origin [future work: point on best fit line of curvatures]

Main findings

- V height implemented via TB height (**expected**)
- V height not implemented via TR backness (**unexpected**)
- V backness implemented via TB height (**unexpected?**)
- V backness implemented via TR backness (**expected**)
- V roundedness implemented via TB and TR (**unexpected**)

Impressionistic:

- 5 levels of TR backness: i > y > ε/œ > α/u > ɔ/u

References

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